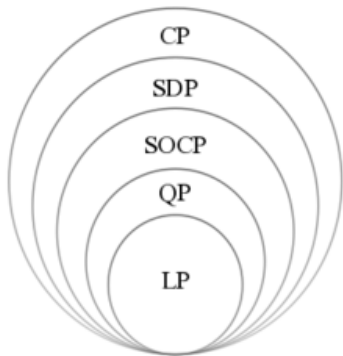
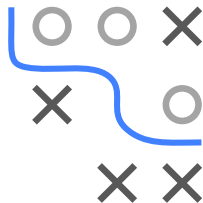


Optimization in Machine Learning

Constrained Optimization

Algorithms for linear programs



Learning goals

- Understand the Simplex algorithm for solving LPs
- Know the two-phase approach for finding a starting point
- Understand how Simplex traverses polytope corners

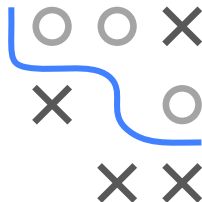
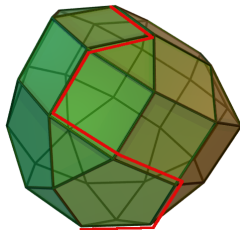
SIMPLEX ALGORITHM

- Most important method for solving linear programs
- Published in 1947 by Georg Dantzig

Basic idea: Start from an arbitrary corner of the polytope.
Run along edges as long as the solution improves. Find a new edge, ...

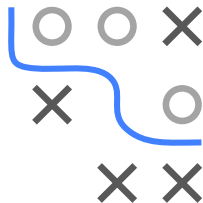
Output: A path along corners of the polytope ending at the optimum

- Since LP is a **convex** optimization problem, the optimal corner found is also a global optimum



SIMPLEX ALGORITHM

- The Simplex algorithm consists of two phases:
 - **Phase I:** Determination of a **starting point**
 - **Phase II:** Determination of the **optimal solution**
- In **Phase I**, a feasible corner $\mathbf{x}_0$ must be found first
- In **Phase II**, this solution is iteratively improved by searching for an edge that improves the objective and running along it to the next corner



SIMPLEX – PHASE I

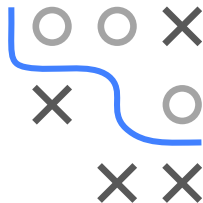
- Find starting point $\mathbf{x}_0$ by solving auxiliary LP with artificial variables ϵ :

$$\min_{\epsilon_1, \dots, \epsilon_m} \sum_{i=1}^m \epsilon_i \quad \text{s.t. } \mathbf{Ax} + \boldsymbol{\epsilon} \geq \mathbf{b}, \quad \epsilon_1, \dots, \epsilon_m \geq 0, \quad \mathbf{x} \geq 0$$

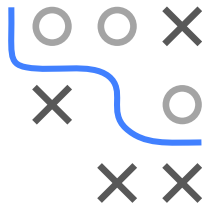
- A feasible starting point for the auxiliary problem is $\mathbf{x} = \mathbf{0}$ and

$$\epsilon_i = \begin{cases} 0 & \text{if } b_i < 0 \\ b_i & \text{if } b_i \geq 0 \end{cases}$$

- We then apply Phase II of Simplex to the auxiliary problem
- If the original problem is feasible, the optimal solution **must** be $\epsilon = (0, \dots, 0)$ (all artificial variables disappear)
- If we find $\epsilon = \mathbf{0}$, we have a valid starting point
- Otherwise, the original problem has no feasible solution



SIMPLEX – EXAMPLE



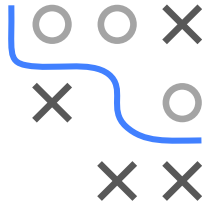
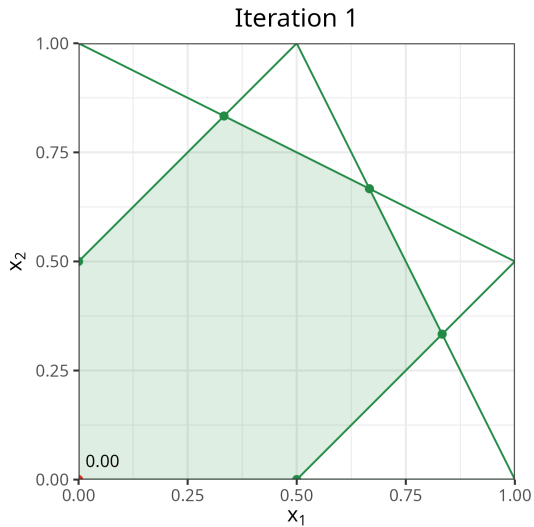
- Consider the following LP:

$$\min_{\mathbf{x} \in \mathbb{R}^2} -x_1 - x_2$$

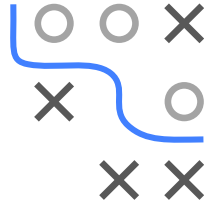
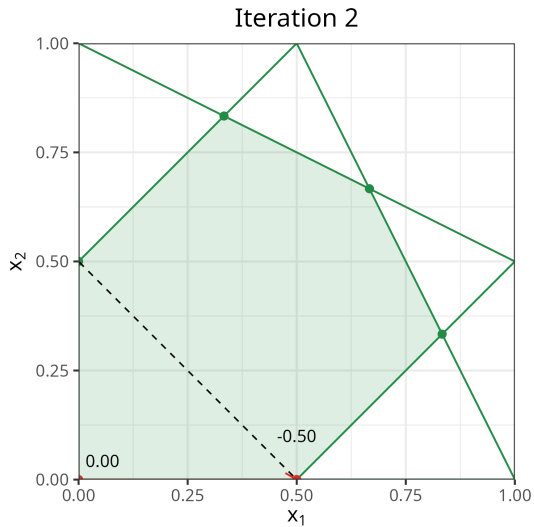
$$\text{s.t. } x_1 - x_2 \geq -0.5, -x_1 - 2x_2 \geq -2, -2x_1 - x_2 \geq -2, -x_1 + x_2 \geq -0.5, \mathbf{x} \geq 0$$

- Starting point is the corner **(0, 0)**

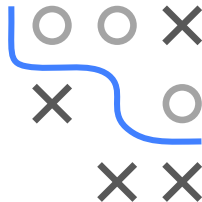
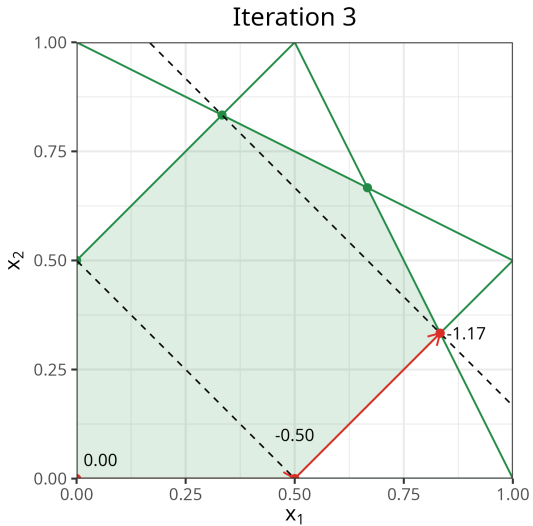
SIMPLEX – EXAMPLE



SIMPLEX – EXAMPLE



SIMPLEX – EXAMPLE



SIMPLEX – EXAMPLE

